# MAT3003 – Probability, Statistics and Reliability

## Tutorial–2 (Module–2)

### Q1. Probability Distribution and Distribution Function

A continuous random variable X has probability density function $f(x)=6x(1-x)$, $0 \le x \le 1$.

(a) Check that f(x) is a probability density function.

(b) Find the cumulative distribution function of X.

(c) Compute $P(X \le 0.5 \mid 0.3 \le X \le 0.7)$.

(d) Determine k such that $P(X<k)=P(X>k)$.

### Q2. Expectation and Variance

The following is the distribution function of a discrete random variable X:

| x | -3 | -1 | 0 | 1 | 2 | 3 | 5 | 8 |
|---|---|---|---|---|---|---|---|---|
| F(x) | 0.10 | 0.30 | 0.45 | 0.65 | 0.75 | 0.90 | 0.95 | 1.00 |

(a) Obtain the probability distribution of X from the given distribution function.

(b) Find P(X is even) and $P(1 \le X < 8)$.

(c) Compute $P(X=-3 \mid X<0)$ and $P(X \ge 3 \mid X>0)$.

(d) Determine the mean and standard deviation of X.

(e) Calculate the mean and variance of $Y=X^2-3X$.

### Q3. Binomial Distribution

When coin 1 is flipped, it lends on heads with probability 0.4: when coin 2 is flipped, it lands on heads with probability 0.7. One of these coins is randomly chosen and flipped 10 times.

(a) Find the probability of exactly 7 heads.

(b) Given that the first flip is a head, find the conditional probability of exactly 7 heads.

### Q4. Poisson Distribution

*The average number of incoming calls at a telephone booth between 10:00 A.M. and 11:00 A.M. is* 4.5 *per two minutes. Assume that the number of phone calls is a Poisson random*

*variable. Determine the probability that the number of incoming calls at this booth during 10:15 A.M. to 10:25 A.M. is*

(a) Find the probability that the number of incoming calls is 30 or fewer.

(b) Find the probability that the number of incoming calls is more than 25.

## Q5. Normal Distribution

*The local authorities in a certain city install* 10,000 *electrical lamps in the streets of the city. If these lamps have an average life of* 1,000 *burning hours with a standard deviation of* 200 *hours, assuming normality, what number of lamps might be expected to fail*

(a) Number of lamps failing in the first 800 hours.

(b) Number of lamps failing between 800 and 1200 hours.

(c) Burning hours after which 10% of the lamps have failed.

(d) Burning hours after which 10% of the lamps are still burning.